

11.1.2 Sum of digits using recursion

ALGORITHM

Step 1: Start

Step 2: Input an integer n

Step 3: Call function $\text{sum}(n)$

Step 4: If $n == 0$, return 0

Step 5: Else compute:

a) $r = n \% 10$ (last digit)

b) $q = n // 10$ (remaining number)

Step 6: Return $r + \text{sum}(q)$

Step 7: Print the result

Step 8: Stop

The screenshot displays the CODETANTRA online coding interface. On the left, the problem statement for "11.1.2. Sum of Digits using Recursion" is shown, including input/output formats and constraints. The main editor on the right contains the following Python code:

```
1 def sum_of_digits_recursive(n):
2
3     if n == 0:
4         return 0
5
6     return (n % 10) + sum_of_digits_recursive(n // 10)
7
8 number = int(input())
```

Below the code editor, performance metrics are shown: Average time 0.011 s, Maximum time 0.019 s. Test results indicate that 2 out of 2 shown test cases and 3 out of 3 hidden test cases passed. A detailed view of Test case 1 shows an expected output of 23 and an actual output of 23 for an input of 5.

